

KEANU REEVES LAURENCE FISHBURNE

THE MATRIX

WARNER BROS. PRESENTS
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A FILM BY LUCY FISHER
KEANU REEVES LAURENCE FISHBURNE "THE MATRIX" CAROL ANNE MORGAN HILARY SWANK JOE PANTOLIANO DON DAVIDS ZACH STAMBERG
COLUMBIA TRI-STAR PICTURES
www.whatisthematrix.com

ON APRIL 2ND THE FIGHT FOR THE FUTURE BEGINS.

Matrix

- A matrix is a collection of numbers ordered by rows and columns into a rectangular array. It is customary to enclose the elements of a matrix in parentheses, brackets, or braces.

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 \\ 3 & 8 \\ -10 & 9 \end{bmatrix}$$

Why do we need matrices?

- A way of representing data (e.g. AFL league table).
- Useful for representing route information.
- Can be used to represent simultaneous equations.

	Played	Won	Drawn	Lost	Goals scored For	Against
Ajax	5	2	1	2	10	5
Battlers	5	2	1	2	4	5
Cloggers	5	2	0	3	7	6
Devils	5	2	0	3	4	11
Enzymes	5	3	2	0	8	2
Flames	5	2	0	3	5	9

5	2	1	2	10	5
5	2	1	2	4	5
5	2	0	3	7	6
5	2	0	3	4	11
5	3	2	0	8	2
5	2	0	3	5	9

Dimensions of a matrix

	5	2	1	2	10	5
ROW	5	2	1	2	4	5
	5	2	0	3	7	6
	5	2	0	3	4	11
	5	3	2	0	8	2
	5	2	0	3	5	9
COLUMN						

This matrix has

SIX rows

SIX columns

It is a SIX by SIX (6 x 6) matrix.

6 x 6 – dimension (or the size of a matrix)

This is also called a **SQUARE** matrix as the **number of rows = number of columns**

More examples

$$\begin{bmatrix} 1 & 0 & 4 \\ 3 & 2 & 0.5 \end{bmatrix}$$

2 rows
and
3 columns.

A 2×3 matrix.

$$\begin{bmatrix} 2 & 5 \\ 11 & -2 \end{bmatrix}$$

2 rows
and
2 columns.

A 2×2 matrix.
(A square matrix.)

$$\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$$

3 rows
and
1 column.

A 3×1 matrix.

COLUMN
MATRIX

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & 3 \end{bmatrix}$$

3 rows
and
4 columns.

A 3×4 matrix.

only 1 column

$$\begin{bmatrix} 5 & 0 & -2 & 1 \end{bmatrix}$$

1 row
and
4 columns

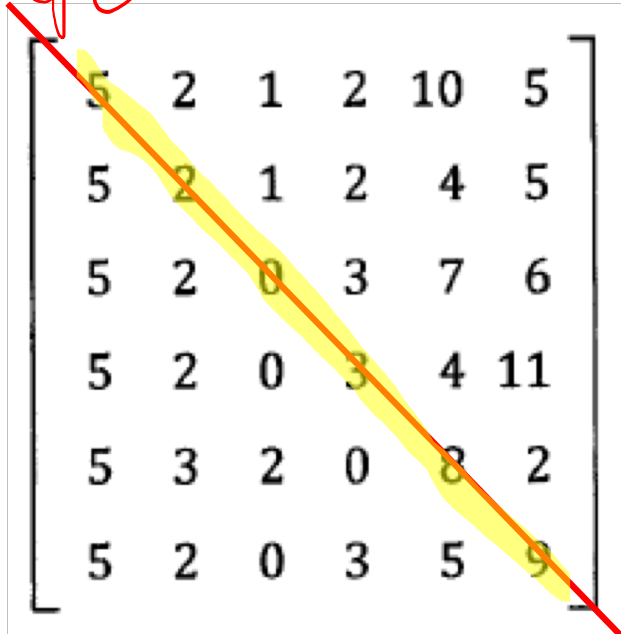
ROW
MATRIX

A 1×4 matrix

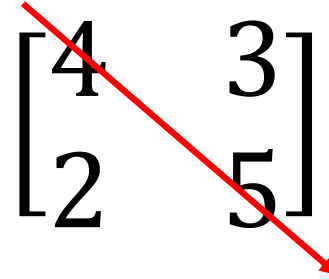
only 1 row

Square Matrix – leading diagonal

Top left



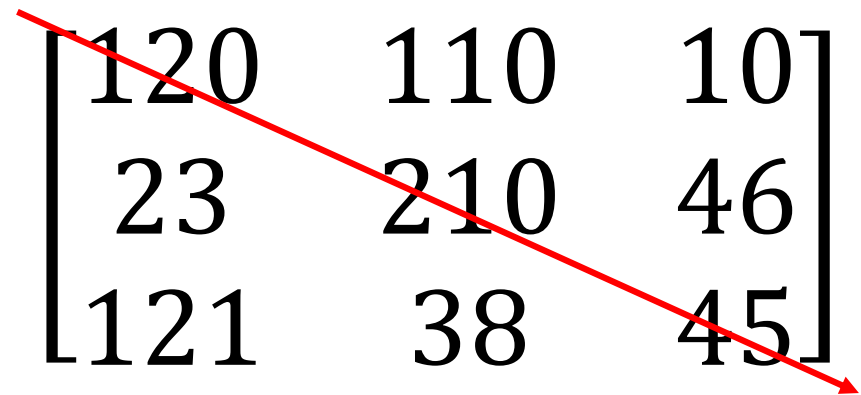
5	2	1	2	10	5
5	2	1	2	4	5
5	2	0	3	7	6
5	2	0	3	4	11
5	3	2	0	8	2
5	2	0	3	5	9



4	3
2	5

Bottom right

ONLY FOR
SQUARE
MATRICES



120	110	10
23	210	46
121	38	45

Square Matrix - Diagonal Matrix

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 120 & 0 & 0 \\ 0 & 21 & 0 \\ 0 & 0 & 145 \end{bmatrix}$$

- All 0 except
the leading
diagonal

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

$$\begin{pmatrix} 4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

5	2	1	2	10	5
5	2	1	2	4	5
5	2	0	3	7	6
5	2	0	3	4	11
5	3	2	0	8	2
5	2	0	3	5	9

Elements of
the matrix

A

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Element is
written as
 $a_{\text{row no.}, \text{column no.}}$

Capital letter to
label a matrix

$$A = \begin{bmatrix} 2 & 0 & -1 & 3 \\ 1 & 6 & -3 & 9 \\ 5 & 7 & 8 & 4 \end{bmatrix}$$

$$a_{12} = 0$$

$$a_{34} =$$

$$a_{21} =$$

$$a_{14} =$$

$$a_{32} =$$

$$a_{23} =$$

$$a_{13} =$$

$$a_{22} =$$

Adding and subtracting matrices

- When adding and subtracting matrices, only matrices of the same dimension can be added or subtracted.

$$A = \begin{bmatrix} 12 & 0 & 5 \\ 2 & 2 & 3 \\ 4 & 6 & 145 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 & 42 \\ 7 & 2 & 5 \\ 3 & 0 & 145 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

Matrix A – 3 x 3 matrix

Matrix B – 3 x 3 matrix

Matrix C – 2 x 2 matrix

$$A = \begin{bmatrix} 120 & 5 & 65 \\ 22 & 21 & 32 \\ 4 & 6 & 145 \end{bmatrix} \quad B = \begin{bmatrix} 20 & 0 & 42 \\ 72 & 21 & 5 \\ 3 & 0 & 145 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 120 + 20 & 5 + 0 & 65 + 42 \\ 22 + 72 & 21 + 21 & 32 + 5 \\ 4 + 3 & 6 + 0 & 145 + 145 \end{bmatrix} = \begin{bmatrix} 140 & 5 & 107 \\ 94 & 42 & 37 \\ 7 & 6 & 290 \end{bmatrix}$$

$$B + A = \begin{bmatrix} 20 + 120 & 0 + 5 & 42 + 65 \\ 72 + 22 & 21 + 21 & 5 + 32 \\ 3 + 4 & 0 + 6 & 145 + 145 \end{bmatrix} = \begin{bmatrix} 140 & 5 & 107 \\ 94 & 42 & 37 \\ 7 & 6 & 290 \end{bmatrix}$$

$$A + B = B + A$$

Therefore addition of matrices is commutative, i.e. the order does not matter.

Since A (or B) and C have different dimensions, addition cannot be performed.

$$A = \begin{bmatrix} 120 & 5 & 65 \\ 22 & 21 & 32 \\ 4 & 6 & 145 \end{bmatrix} \quad B = \begin{bmatrix} 20 & 0 & 42 \\ 72 & 21 & 5 \\ 3 & 0 & 145 \end{bmatrix} \quad C = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 120 - 20 & 5 - 0 & 65 - 42 \\ 22 - 72 & 21 - 21 & 32 - 5 \\ 4 - 3 & 6 - 0 & 145 - 145 \end{bmatrix} = \begin{bmatrix} 100 & 5 & 23 \\ -50 & 0 & 27 \\ 1 & 6 & 0 \end{bmatrix}$$

$$B - A = \begin{bmatrix} 20 - 120 & 0 - 5 & 42 - 65 \\ 72 - 22 & 21 - 21 & 5 - 32 \\ 3 - 4 & 0 - 6 & 145 - 145 \end{bmatrix} = \begin{bmatrix} -100 & -5 & -23 \\ 50 & 0 & -27 \\ -1 & -6 & 0 \end{bmatrix}$$

$$A - B \neq B - A$$

Therefore subtraction of matrices is **NOT** commutative, i.e. the order is important.

Since A (or B) and C have different dimensions, subtraction cannot be performed.

Scalar multiplication

$$C = \begin{bmatrix} 4 & 7 & 12 \\ 6 & 9 & 16 \end{bmatrix} \qquad D = \begin{bmatrix} 2 & 12 \\ 10 & 8 \\ 4 & 7 \end{bmatrix}$$

$$2C = \begin{bmatrix} 2 \times 4 & 2 \times 7 & 2 \times 12 \\ 2 \times 6 & 2 \times 9 & 2 \times 16 \end{bmatrix} = \begin{bmatrix} 8 & 14 & 24 \\ 12 & 18 & 32 \end{bmatrix}$$

$$0.5D = \begin{bmatrix} 0.5 \times 2 & 0.5 \times 12 \\ 0.5 \times 10 & 0.5 \times 8 \\ 0.5 \times 4 & 0.5 \times 7 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 5 & 4 \\ 2 & 3.5 \end{bmatrix}$$

If $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 & -2 \\ 1 & 0 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$ determine each of the following. If any cannot be determined state this clearly and give the reason.

(a) $A + B$

(b) $A + C$

(c) $B - A$

(d) $5C$

(e) $3B - 2A$

$$\begin{aligned} a) \quad A + B &= \begin{pmatrix} 1 & 2 & 4 \\ 0 & -4 & 5 \end{pmatrix} + \begin{pmatrix} 3 & 5 & -2 \\ 1 & 0 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 7 & 2 \\ 1 & -4 & 3 \end{pmatrix} \end{aligned}$$

b) $A + C$ cannot be determined because they have different dimensions

$$c) \quad B - A = \begin{pmatrix} 2 & 3 & -6 \\ 1 & 4 & -7 \end{pmatrix}$$

$$d) \quad 5C = \begin{pmatrix} 10 & 15 \\ 5 & -25 \end{pmatrix}$$

$$\begin{aligned} e) \quad 3B - 2A &= \begin{pmatrix} 9 & 15 & -6 \\ 3 & 0 & -6 \end{pmatrix} - \begin{pmatrix} 2 & 4 & 8 \\ 0 & -8 & 10 \end{pmatrix} \\ &= \begin{pmatrix} 7 & 11 & -14 \\ 3 & 8 & -16 \end{pmatrix} \end{aligned}$$

Suppose that the 3×2 matrix shown on the right shows the cost of three models of gas heater in two different shops.

	Shop One	Shop Two
Economy	\$250	\$280
Standard	\$340	\$330
Deluxe	\$450	\$450

Now suppose that in a sale both shops offer 10% discount on all models.

↓
New price = 90% of the original price

$$\begin{aligned} \text{New Price} &= 90\% \begin{pmatrix} 250 & 280 \\ 340 & 330 \\ 450 & 450 \end{pmatrix} \\ &= \begin{pmatrix} \$225 & \$252 \\ \$306 & \$297 \\ \$405 & \$405 \end{pmatrix} \end{aligned}$$

Equal matrices

$$\text{If } \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} = \begin{bmatrix} 3 & 11 \\ 13 & 23 \\ 18 & 4 \end{bmatrix} \text{ then}$$

$$a = 3, b = 13, c = 18, d = 11, e = 23, f = 4$$